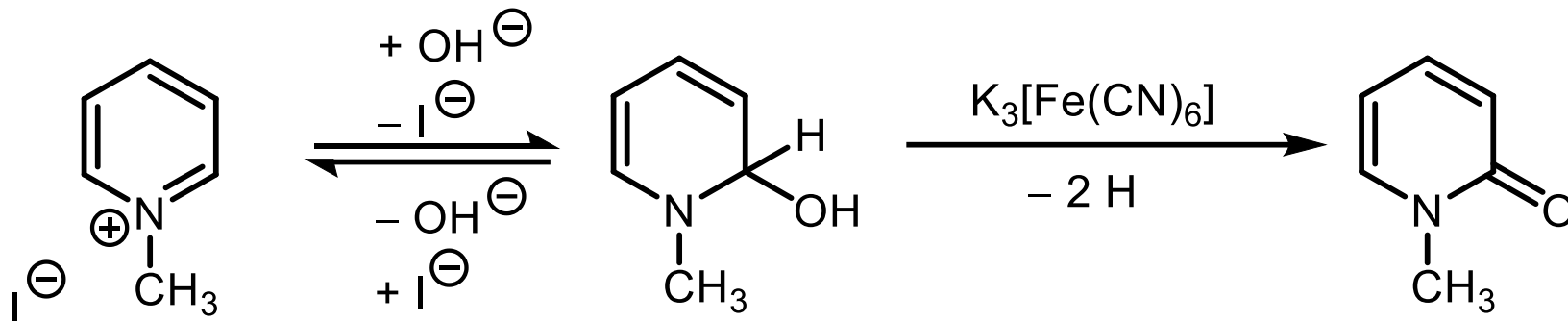
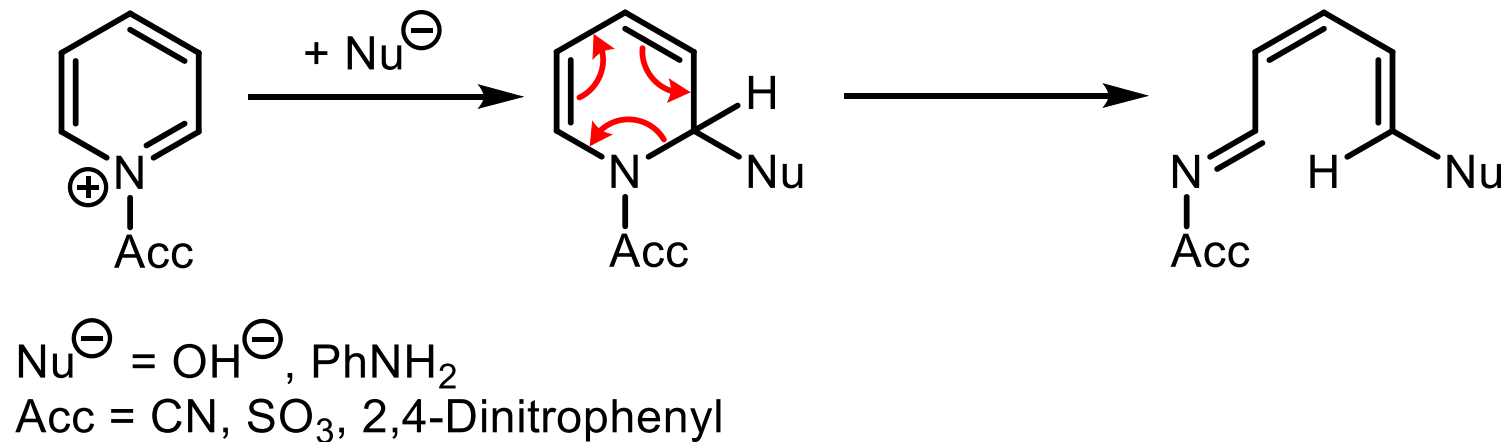


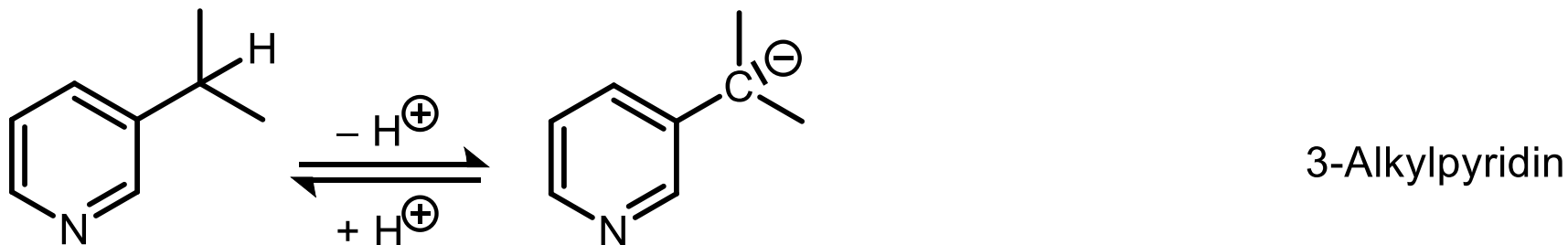
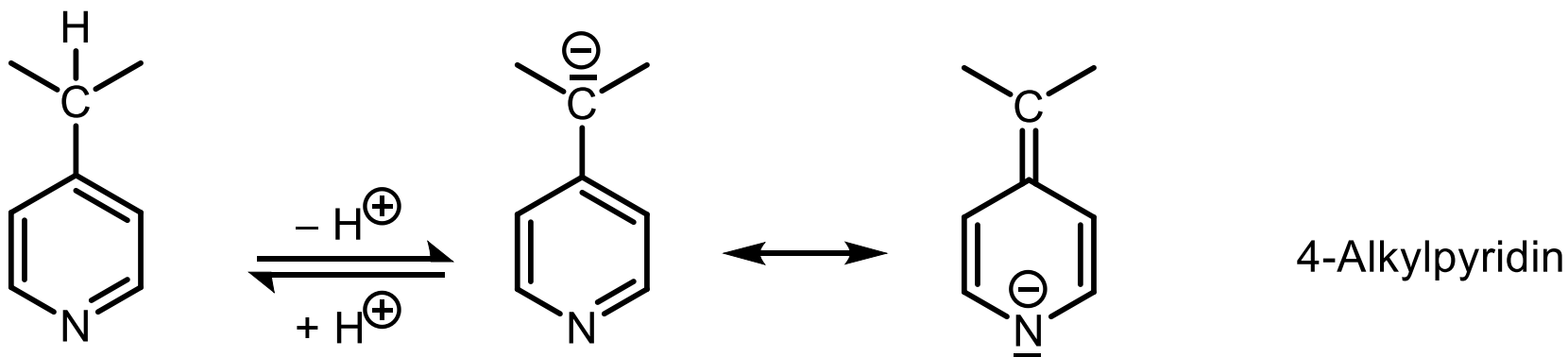
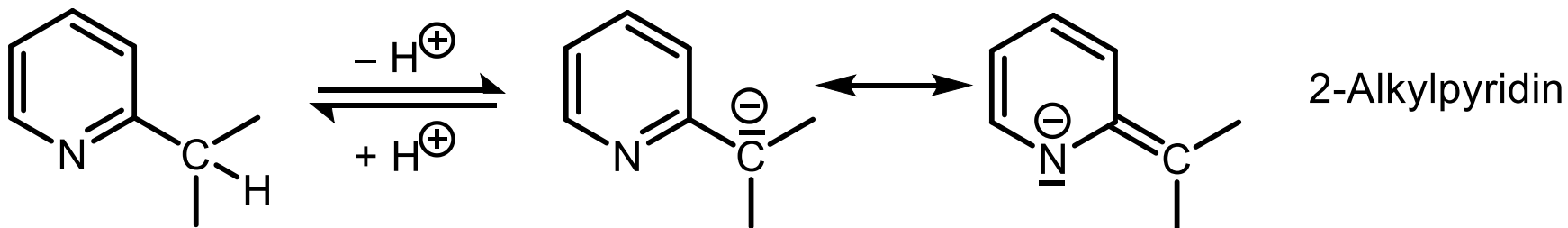
4) Nucleophile Addition an Pyridiniumionen



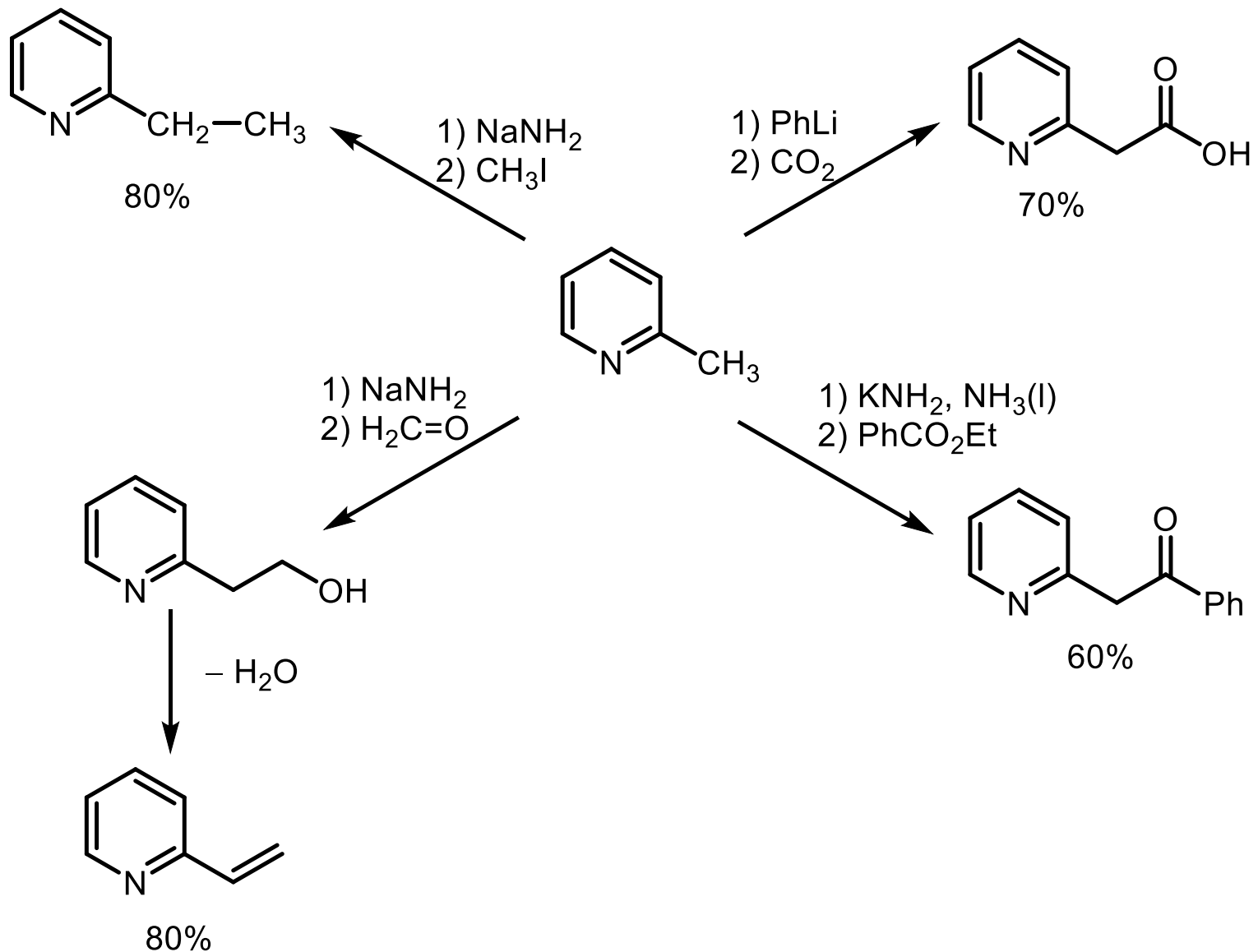
Zincke-Reaktion



5) Seitenkettenreaktivität an Pyridinen

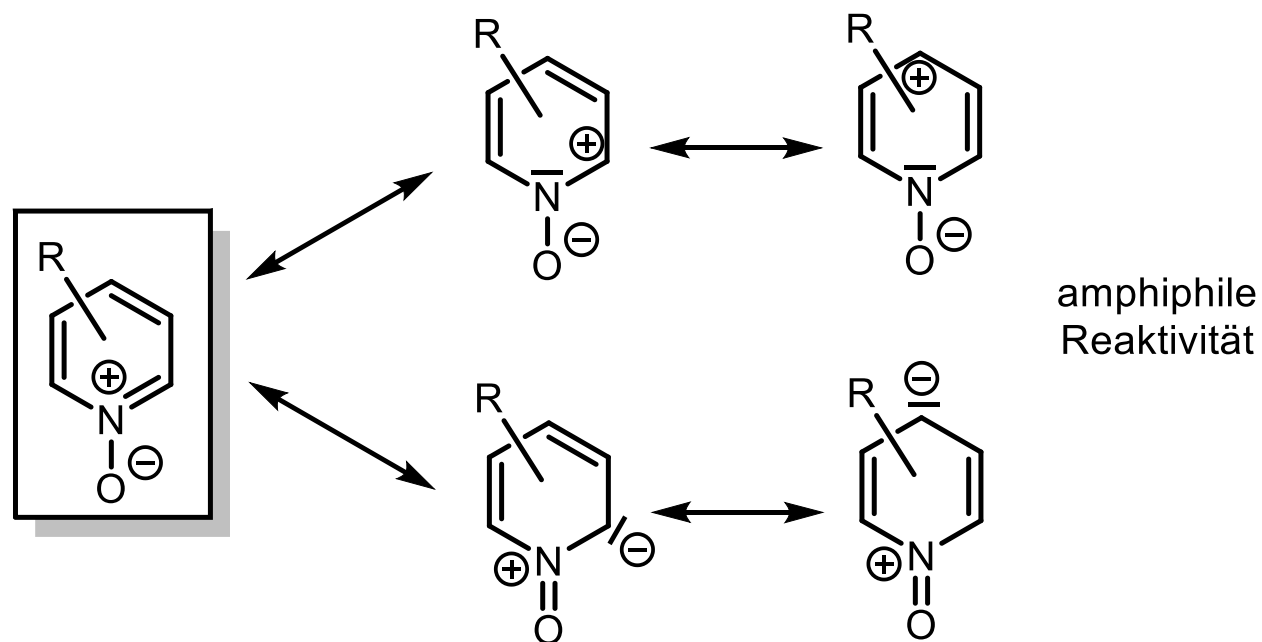
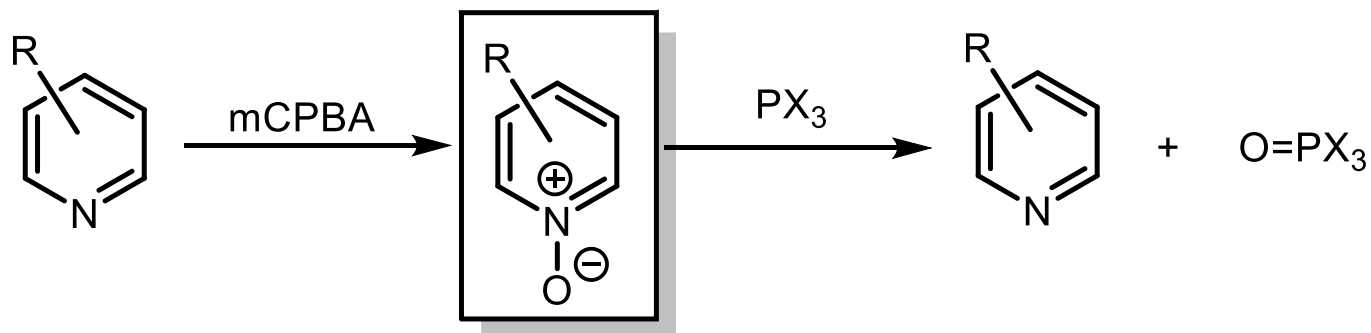


5.1.1.2. Reaktivität Seitenkettenacidität

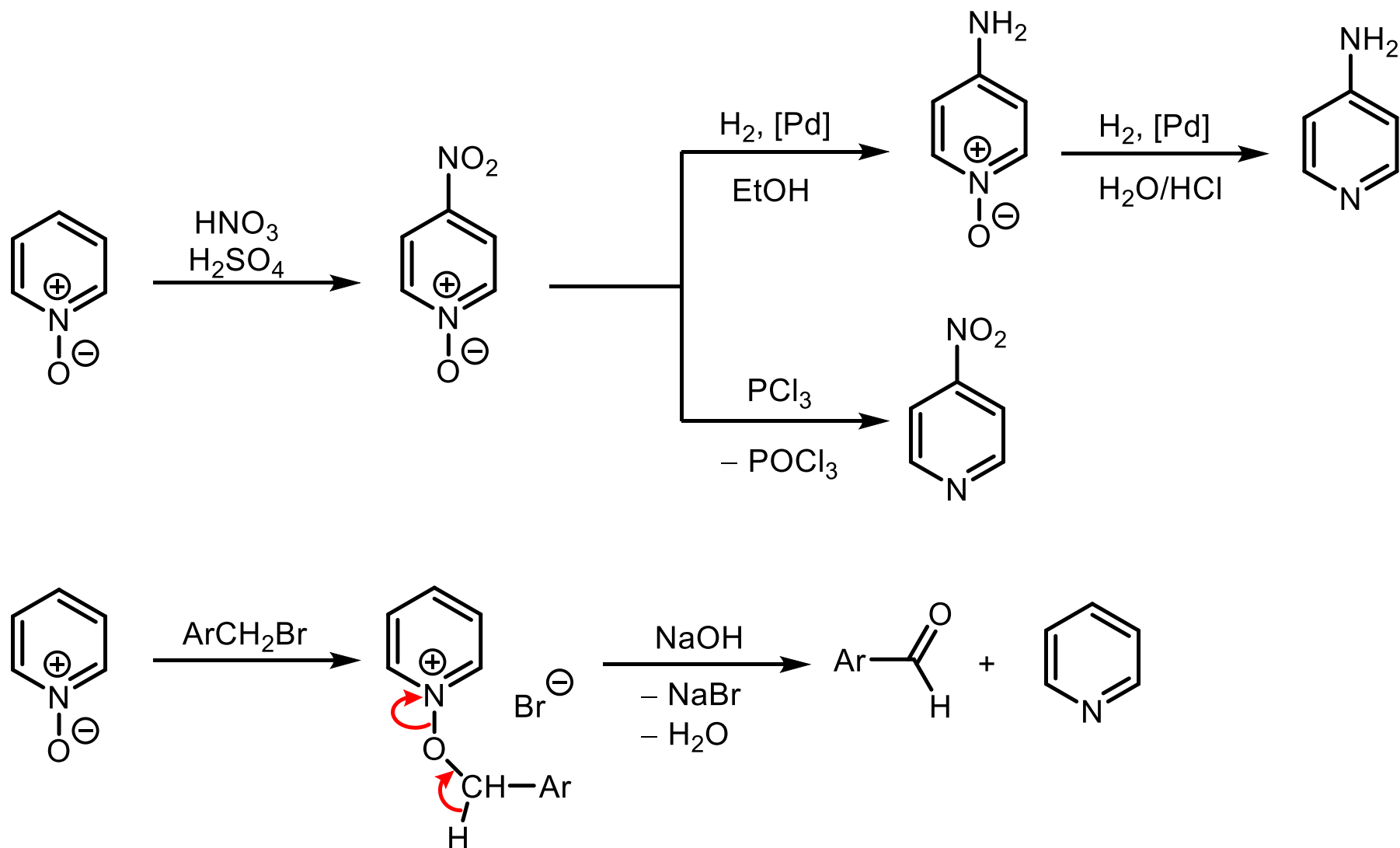


6) Reaktionen von Pyridin-N-Oxid

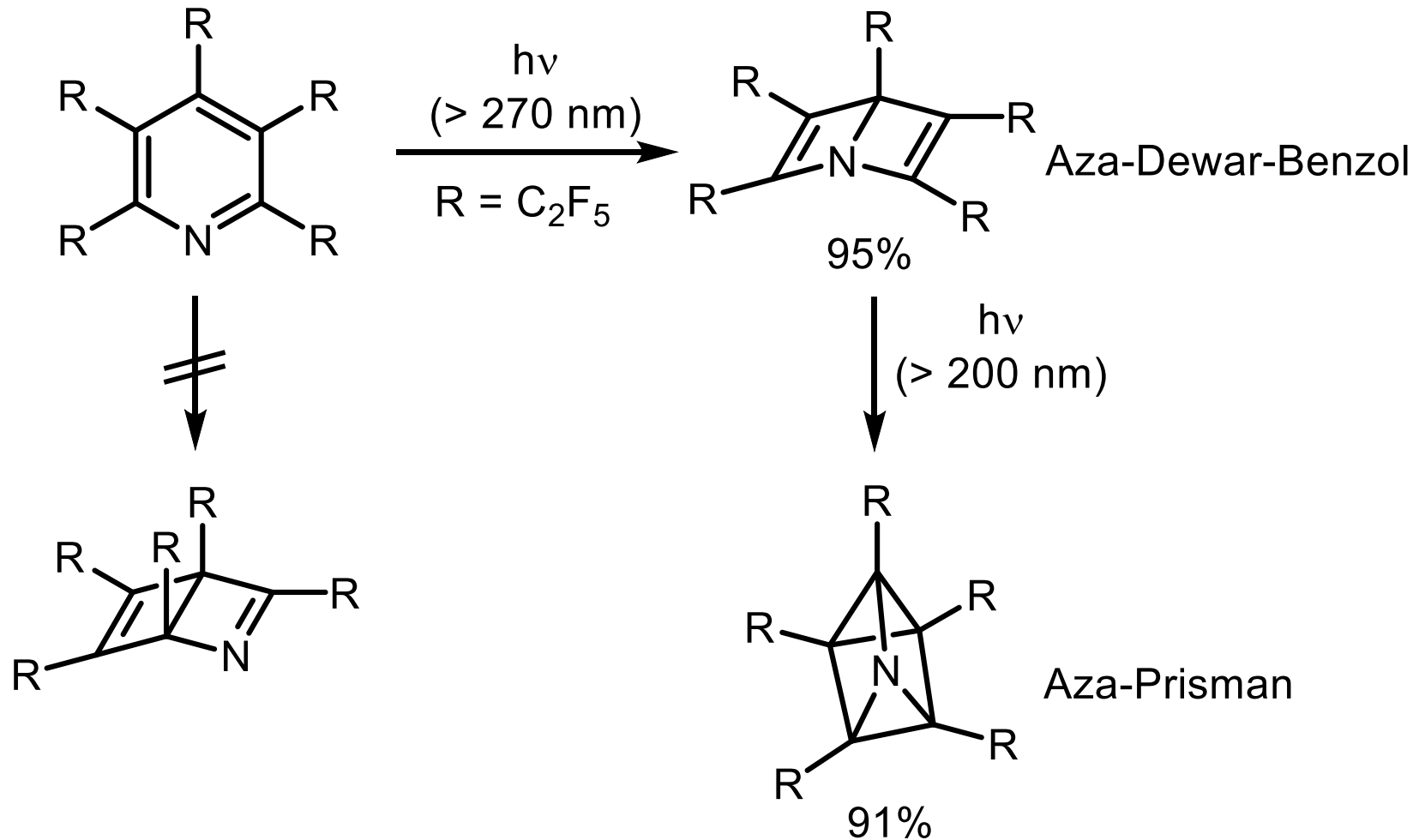
86



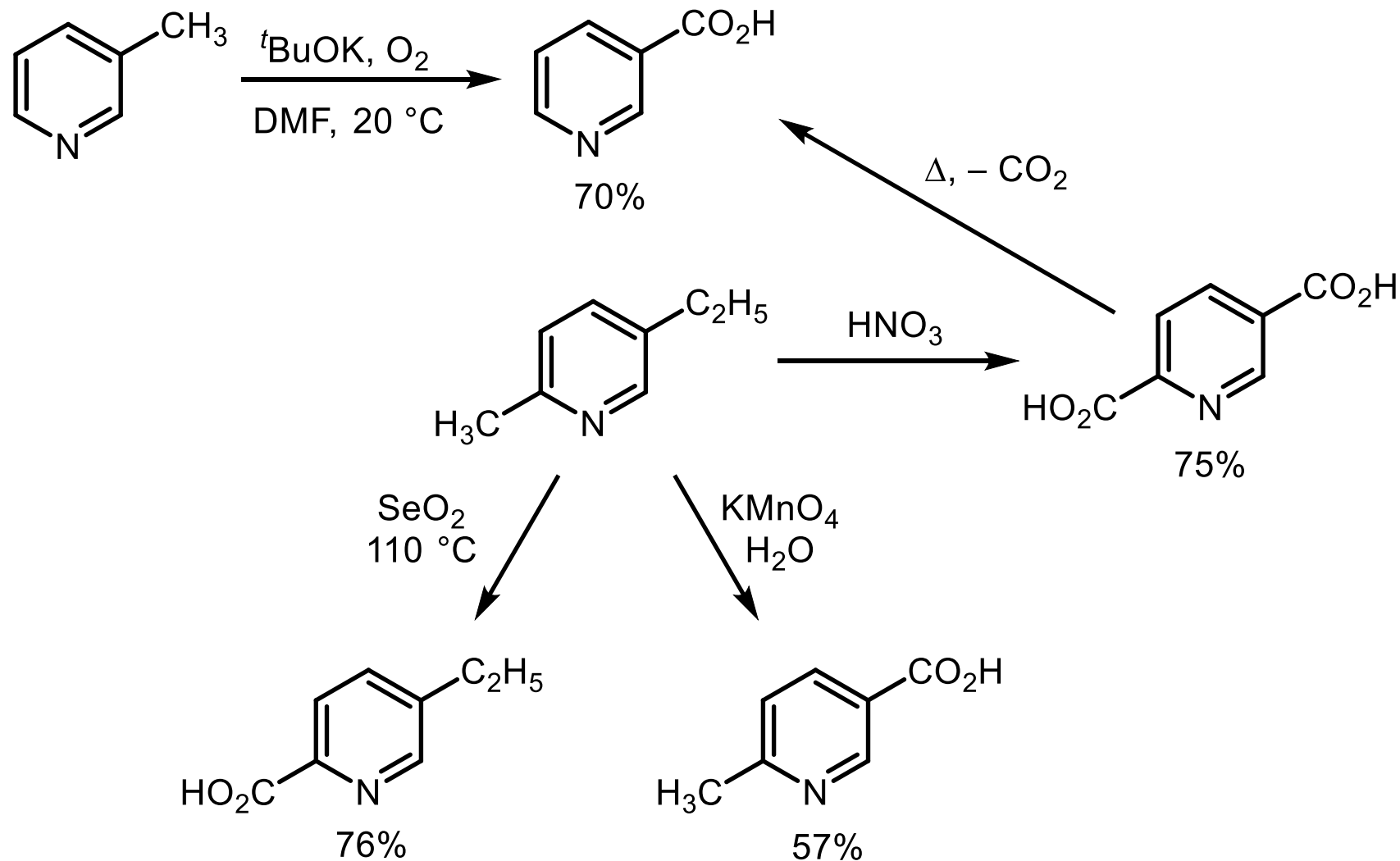
5.1.1.2. Reaktivität Pyridin-N-Oxid



7) Photochemische Reaktionen von Pyridin



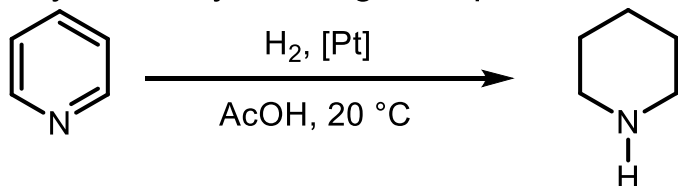
8) Oxidationsreaktionen



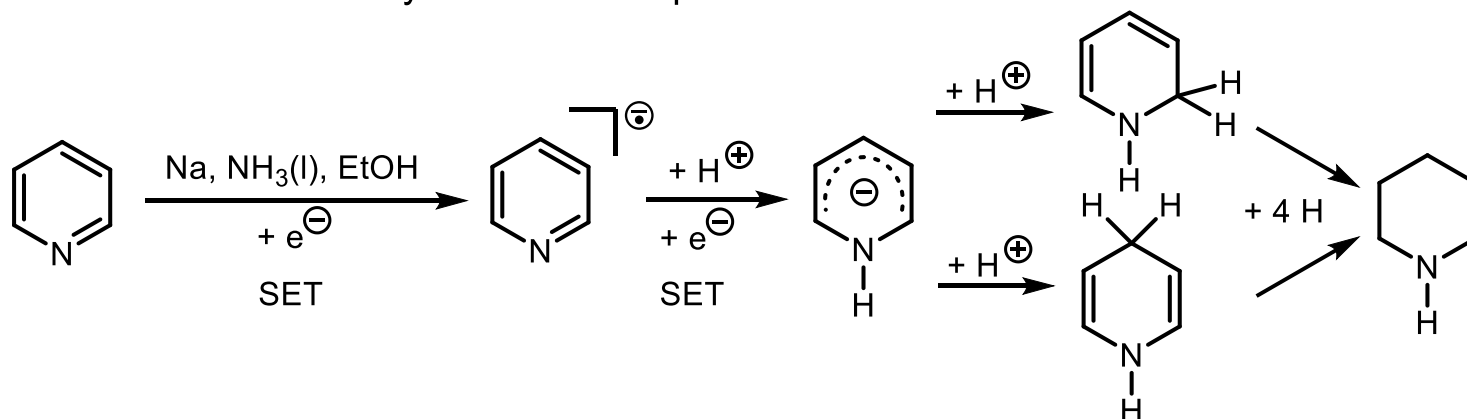
5.1.1.2. Reaktivität Redoxreaktionen

9) Reduktionsreaktionen

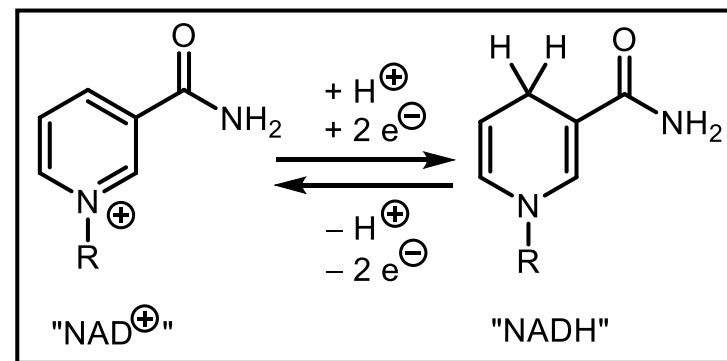
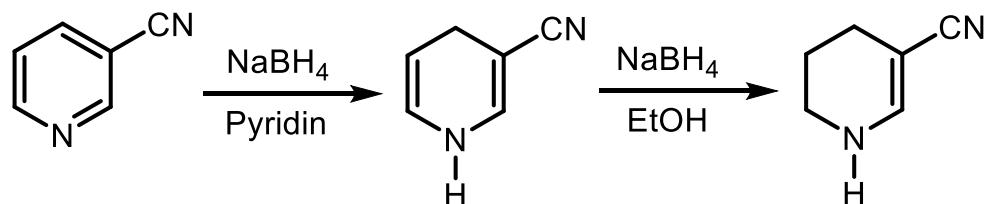
Katalytische Hydrierung zu Piperidin



Birch-Reduktion von Pyridin mit Na in protischem Medium

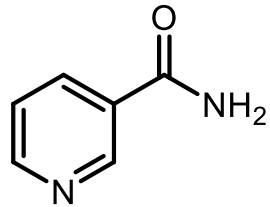


Borant-Reduktion von Acceptor-Pyridinen

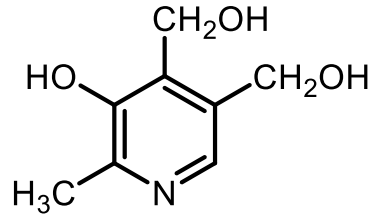


5.1.1.3. Vorkommen und Bedeutung

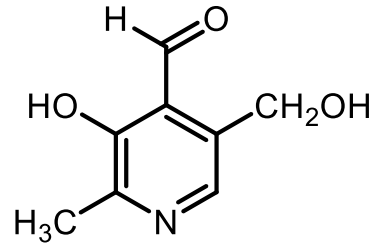
1) Naturstoffe



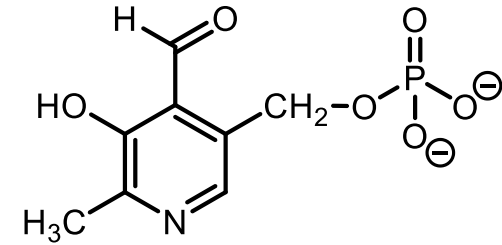
Nicotinamid
(Niacin, Vitamin B₅)



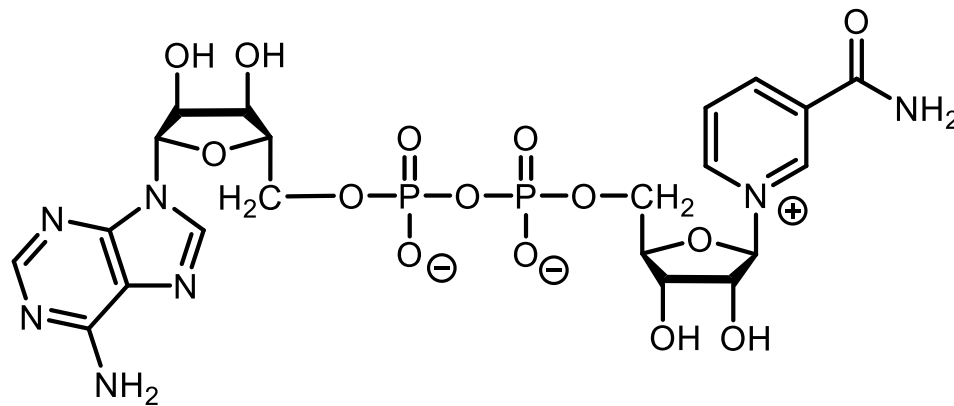
Pyridoxol
(Pyridoxin, Vitamin B₆)



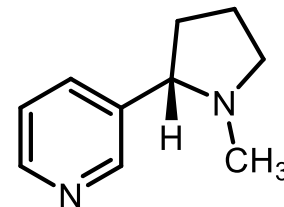
Pyridoxal



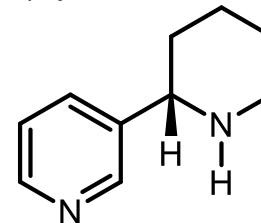
Pyridoxalphosphat
(Cofaktor der Transaminasen;
Aminosäurestoffwechsel)



NAD⁺
(Nicotinamidadenindinucleotid)



Nicotin
(Pyridinalkaloid)

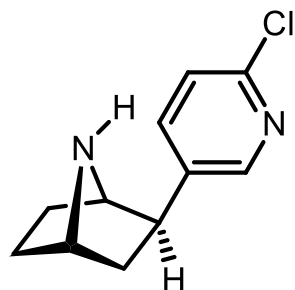


Anabasin
(Pyridinalkaloid)



Nicotin:
Hauptalkaloid der Tabakpflanze *Nicotiana tabacum* wirkt als Acetylcholin-Agonist, also als Neurotransmitter, auf die Nicotin-Rezeptoren. Tödliche Dosis: 100 mg/Mensch.

5.1.1.3. Vorkommen und Bedeutung Natur- und Wirkstoffe

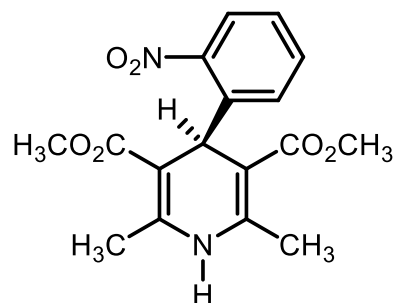


Epibatidin

Epibatidin wird von südamerikanischen Pfeilgiftfröschen mit der Nahrung aufgenommen. 200-fach analgetische Wirkung von Morphin, allerdings als Acetylcholin-Agonist auf die Nicotin-Rezeptoren. Isolierung 1992 durch Daly et al., Synthese z. B. durch Corey et al. (1993).®

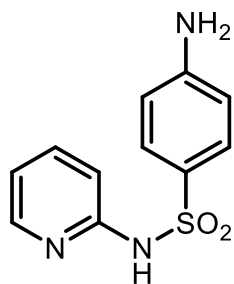


2) Wirkstoffe



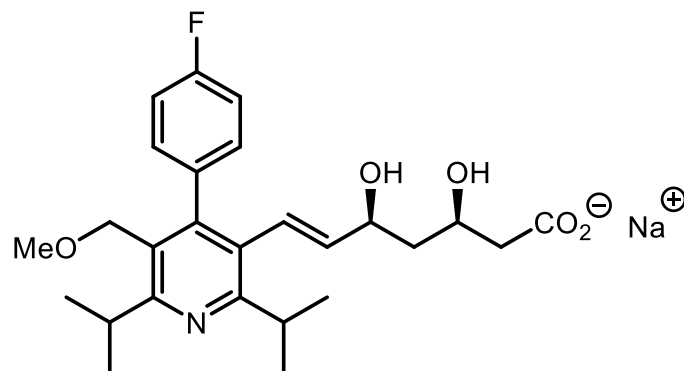
Nifedipin

(Coronar-Therapeutikum)



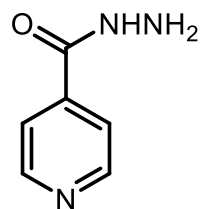
Sulfapyridin

(Sulfonamid-Antibiotikum)



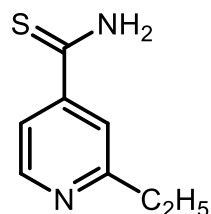
Cervivastatin (Lipobay®)

(Cholesterinspiegel-Senker)

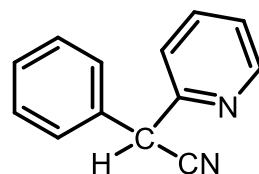


Isoniazid

(Tuberculostatika)



Ethionamid

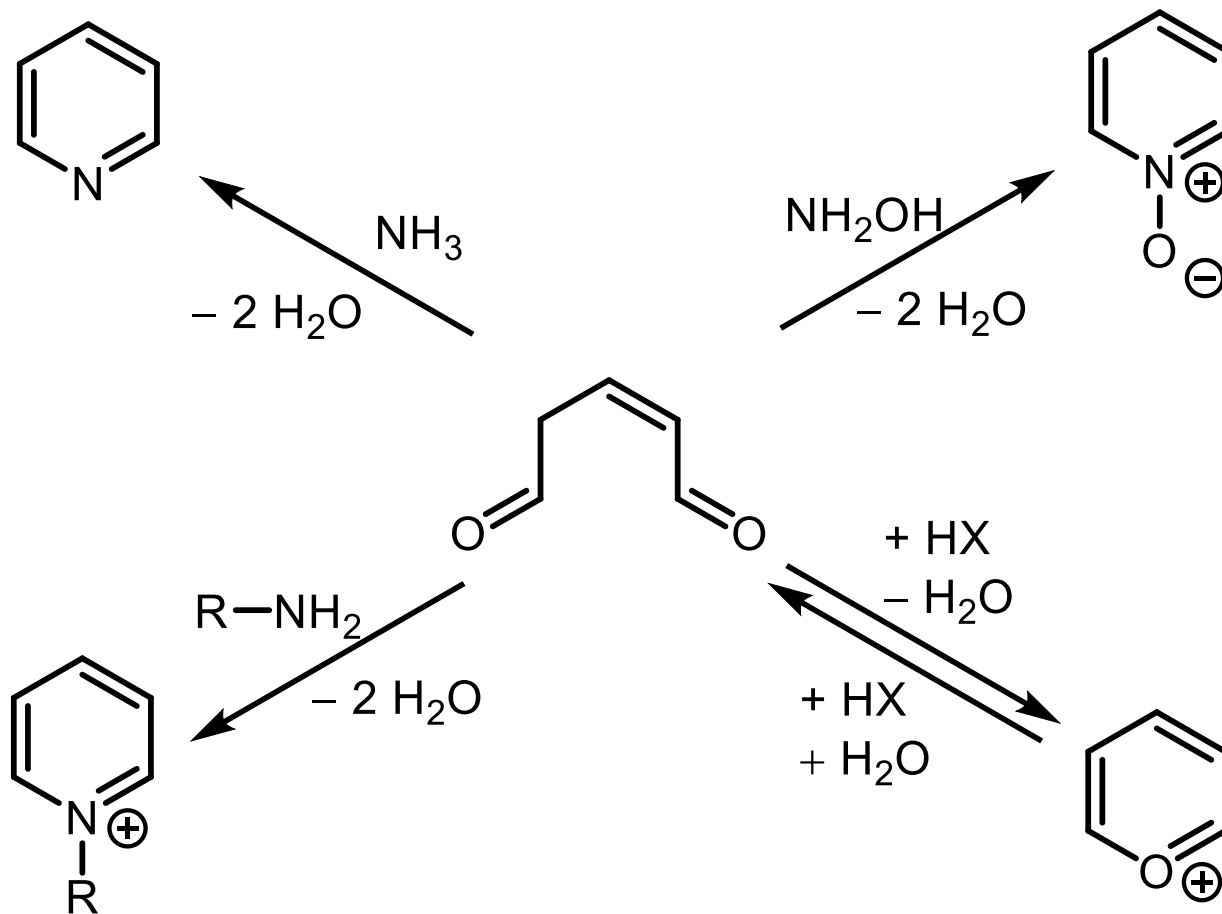


Pheniramin

(Antihistaminikum)

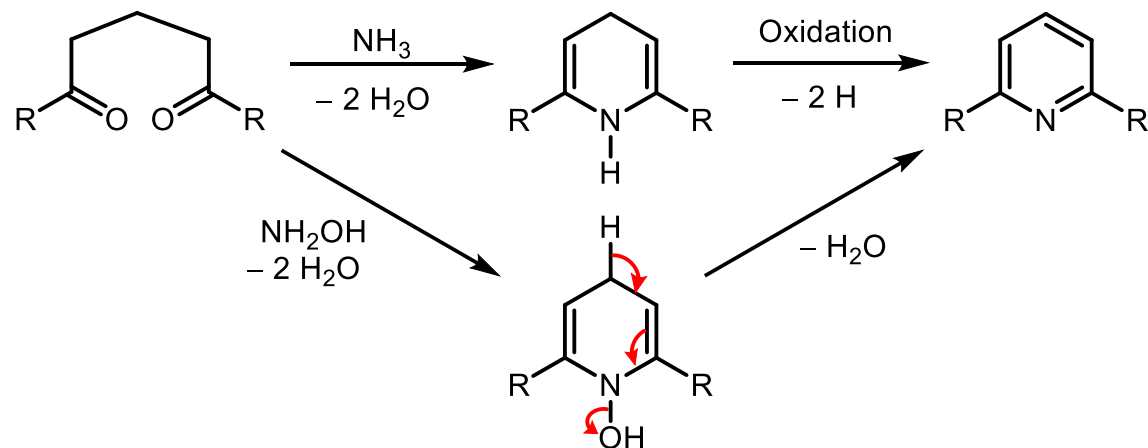
5.1.1.4. Ringaufbauende Synthesen

1) Aus Glutacondialdehyd (Cyclokondensation)



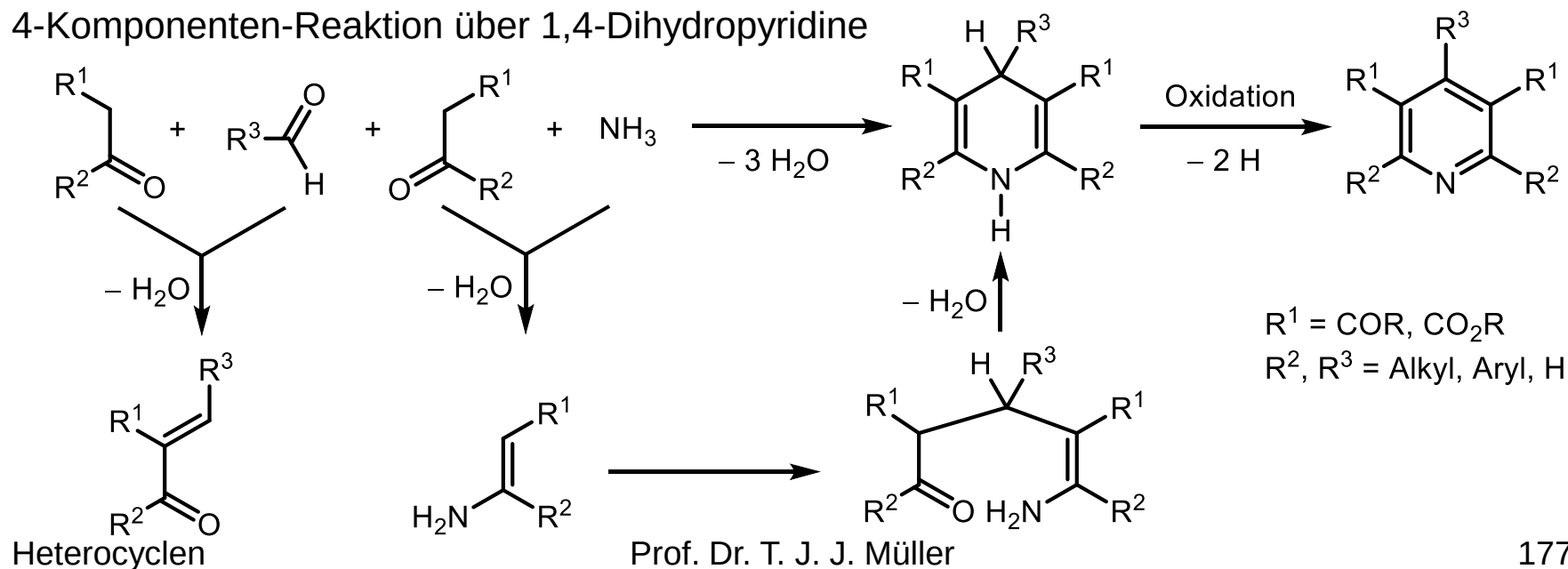
5.1.1.4. Ringaufbauende Synthesen Cyclokondensationen

2) Aus 1,5-Dicarbonyl-Verbindungen und Ammoniak



3) Hantzsch-Synthese von Pyridinen

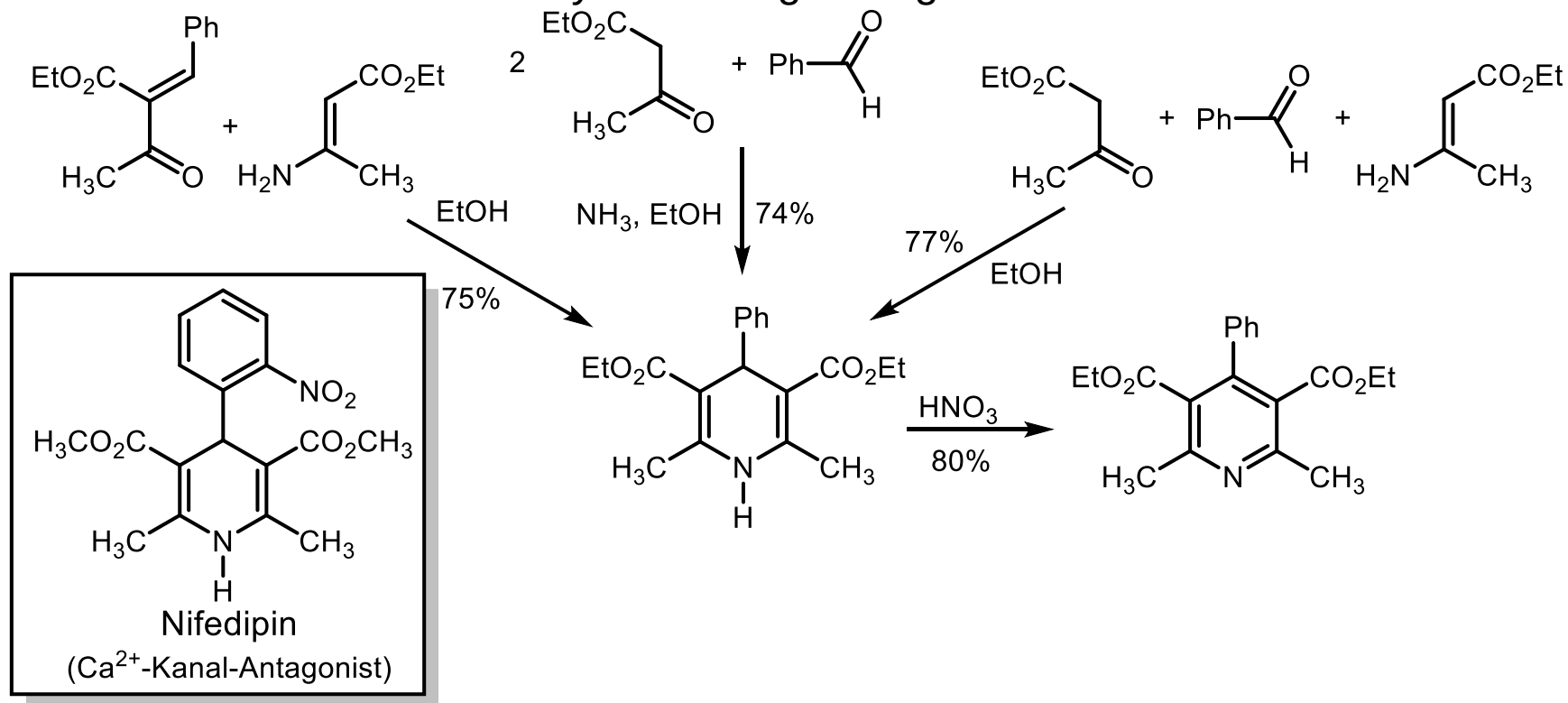
4-Komponenten-Reaktion über 1,4-Dihydropyridine



5.1.1.4. Ringaufbauende Synthesen

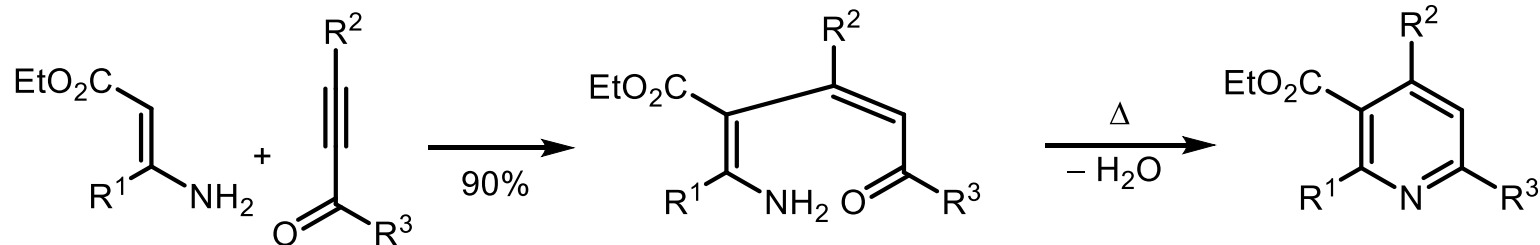
Cyclokondensationen

Es sind mehrere "Hantzsch-Synthesewege" möglich!



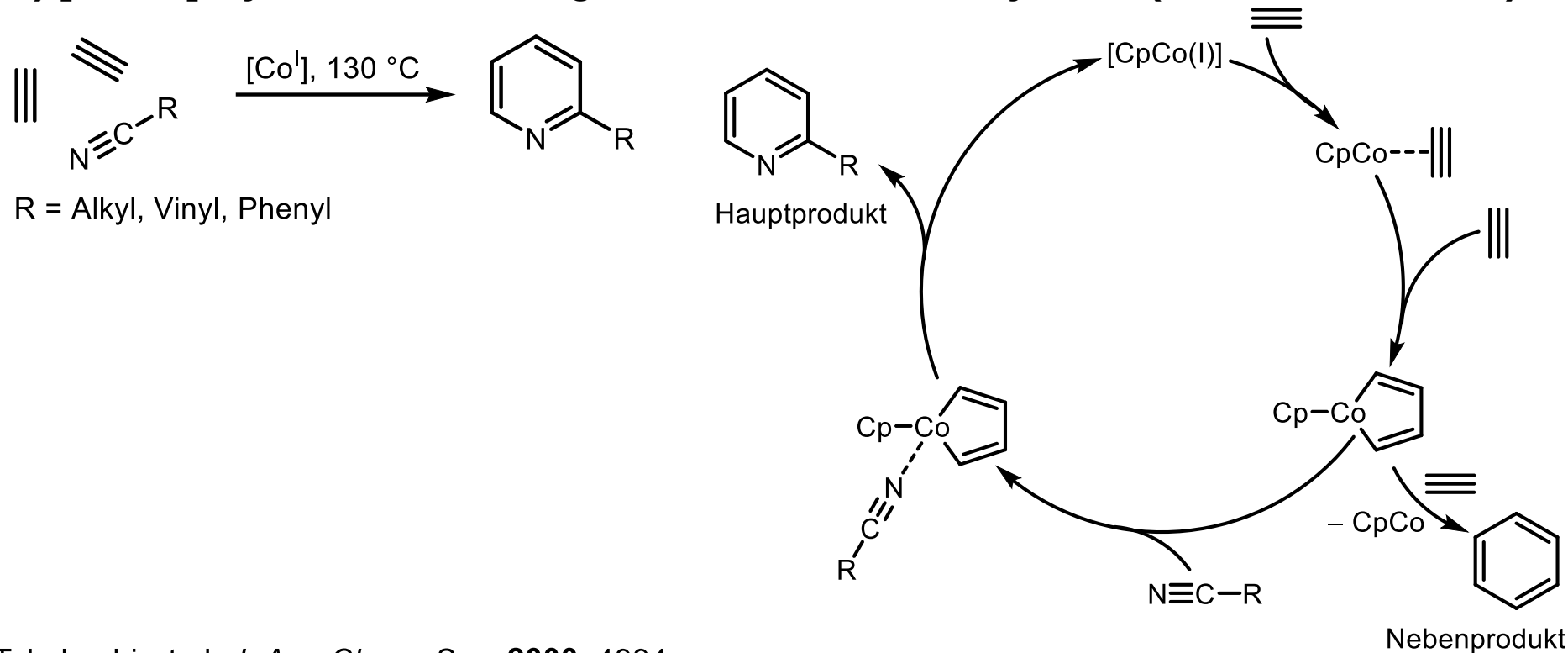
4) Bohlmann-Rahtz-Synthese (1957)

Michael-Addition
von Enaminen
an Alkinone

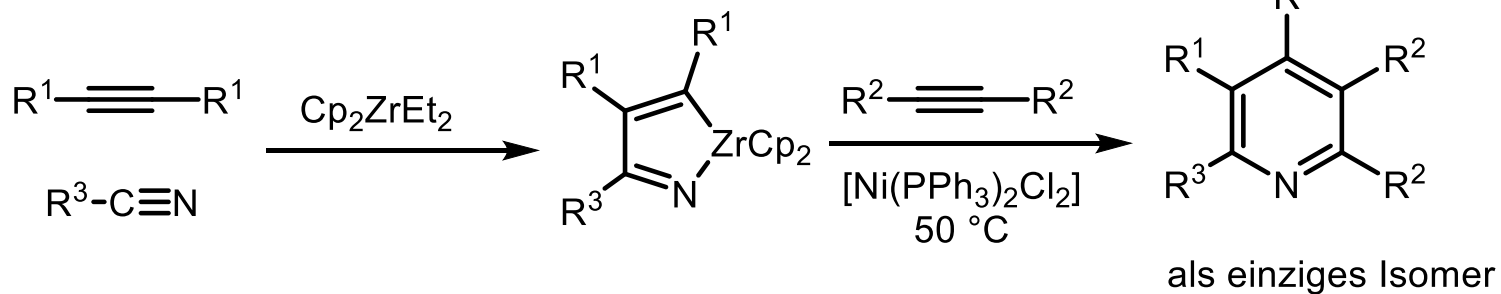


5.1.1.4. Ringaufbauende Synthesen Metallorganische Synthesen

5) [2+2+2]-Cyclotrimerisierung von Nitrilen und Acetylenen (Bönnemann, 1978)



Takahashi et al. *J. Am. Chem. Soc.* **2000**, 4994.



Heterocyclen

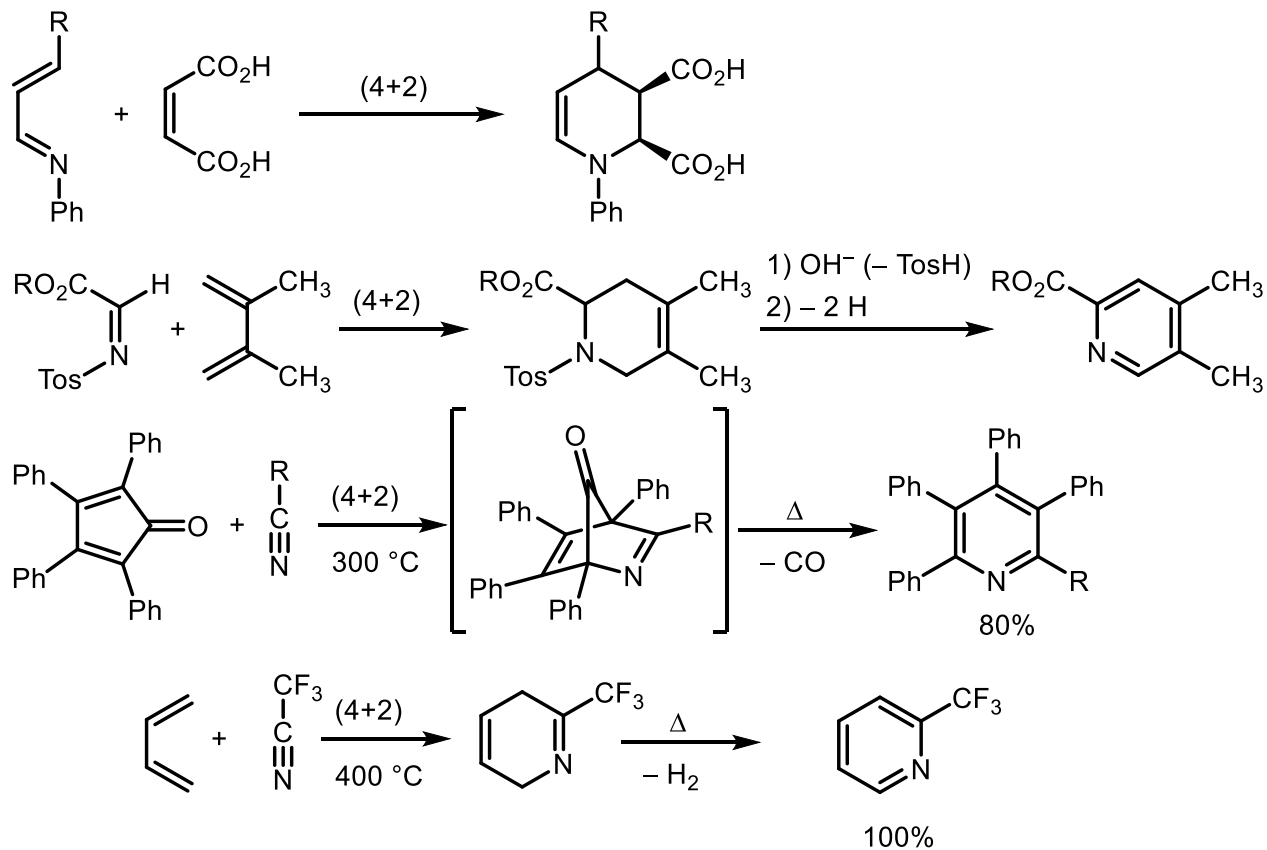
Prof. Dr. T. J. J. Müller

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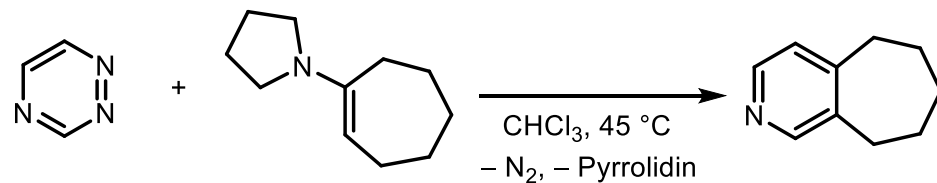
5.1.1.4. Ringaufbauende Synthesen

Cycloadditionen

6) Über [4+2]-Cycloadditionen

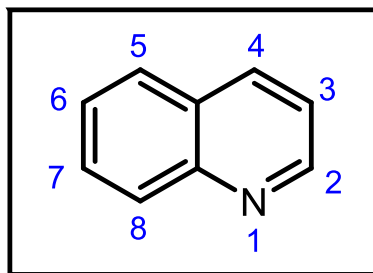


Neunhöffer-Boger-Synthese



5.1.2. Chinolin (engl. Quinoline)

5.1.2.1. Struktur, Eigenschaften, Vorkommen, Bedeutung

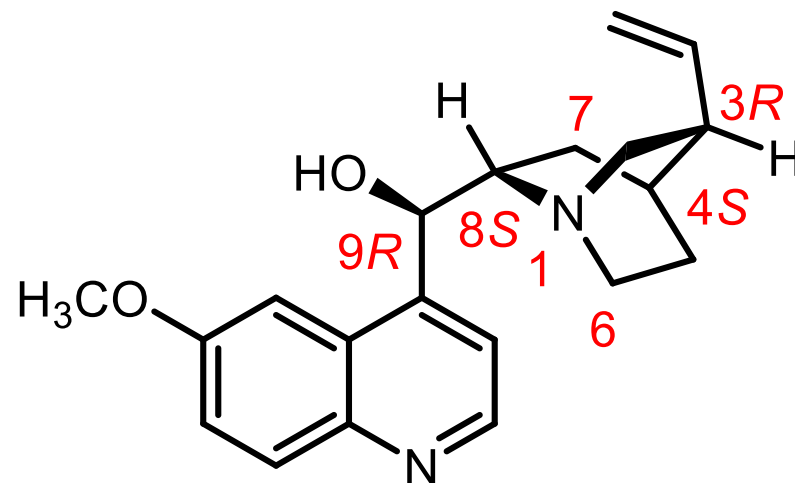


farblose, wasserdampfflüchtige Flüssigkeit mit charakteristischem Geruch; Sdp.: 237 °C

Im (-)-Chinin (engl. quinine; Analogon ohne Methoxygruppe: Cinchonin) aus der Rinde subtropischer Bäume der Gattungen *Cinchona* und *Remija*, isoliert 1820 von Pelletier und Caventou;

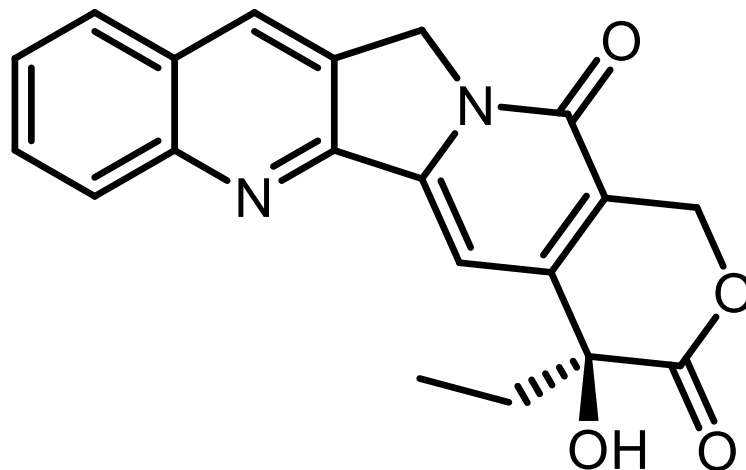
Eingesetzt als Chemotherapeutikum gegen Malaria
Konstitutionsbeweis durch Totalsynthese (Woodward, Doering 1944).

Hemmung des Wachstums der ungeschlechtlichen Formen des Malaria-Erregers *Plasmodium falciparum*.
Chinidin: 8*R*,9*S*-Diastereomer von Chinin (Analogon ohne Methoxygruppe: Cinchonidin).



5.1.2.1. Struktur, Eigenschaften, Vorkommen, Bedeutung (Chinolin)

Camptothecin

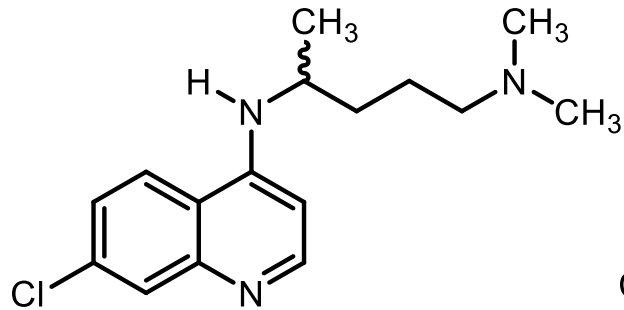


aus *Camptotheca acuminata* (Chinesischer Glücksbaum)

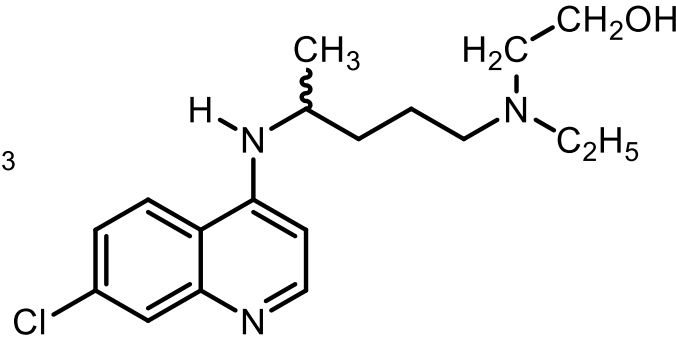
DNA-Topoisomerase I-Inhibitor, seit 1996 zugelassen zur Therapie verschiedener Krebsarten

5.1.2.1. Struktur, Eigenschaften, Vorkommen, Bedeutung (Chinolin)

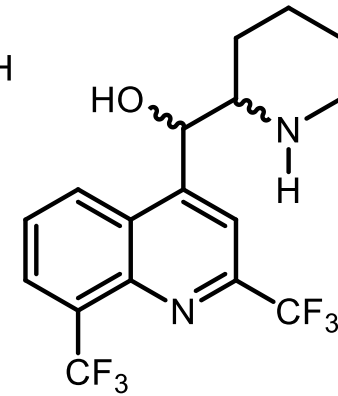
Pharmaka



Chloroquin
(Resochin)

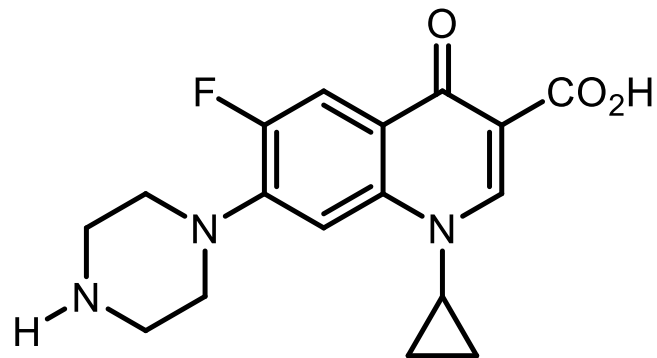


Hydroxychloroquin
(potenziell für die Behandlung von Covid-19)



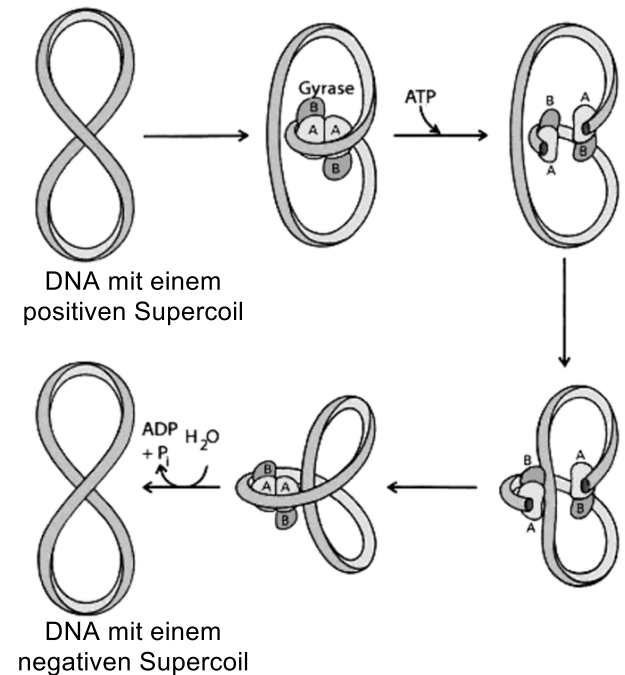
Mefloquin
(Antiparasitikum)

Chinin-Mimetika, wirksam gegen Schizonten des Malaria-Erregers *Plasmodium*



Ciprofloxacin
(Ciprobay[®], antibakterieller
Gyrase-Inhibitor)

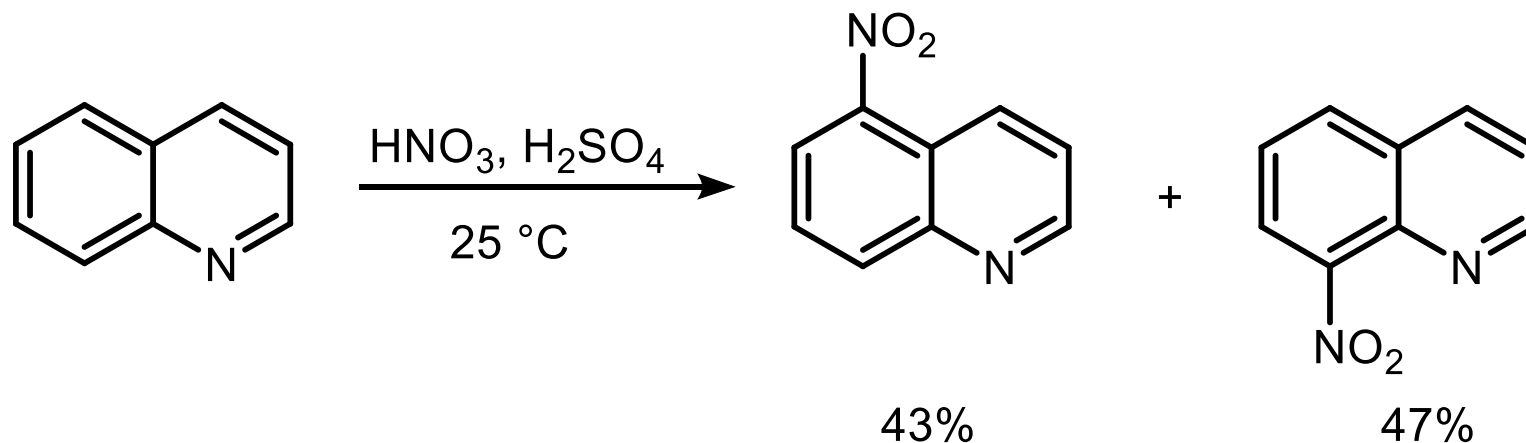
Gyrase: eine DNA-Topoisomerase II, verantwortlich für das negative "Supercoiling" der DNA und die Packung der DNA während der Replikation.



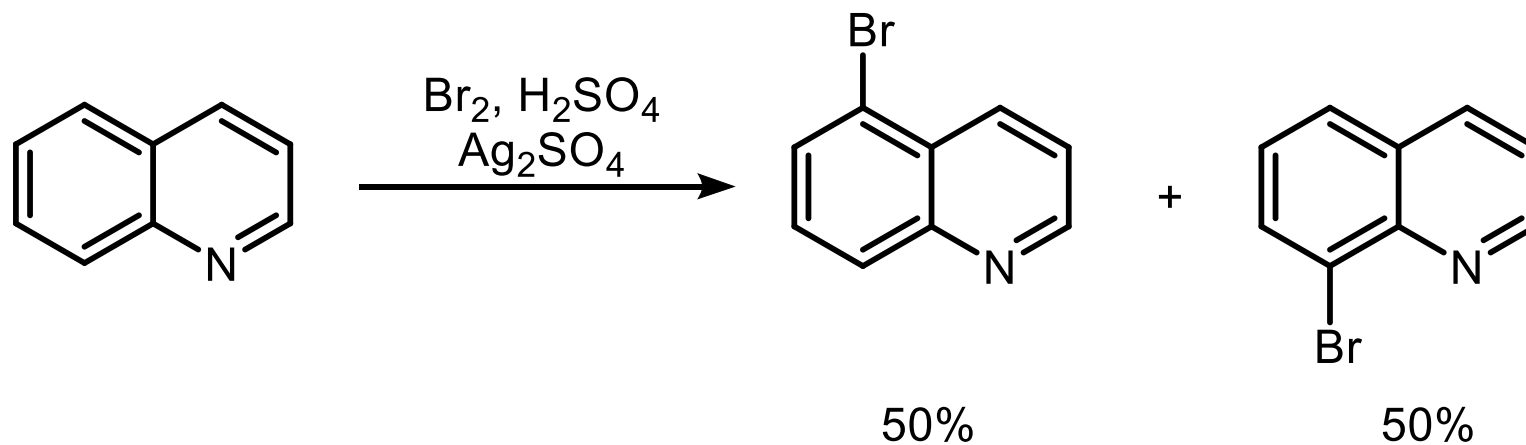
5.1.2.2. Reaktionen

1) Elektrophile Substitutionsreaktionen (S_EAr): C-8 > C-5/C-6 > C-7 > C-3

Nitrierung

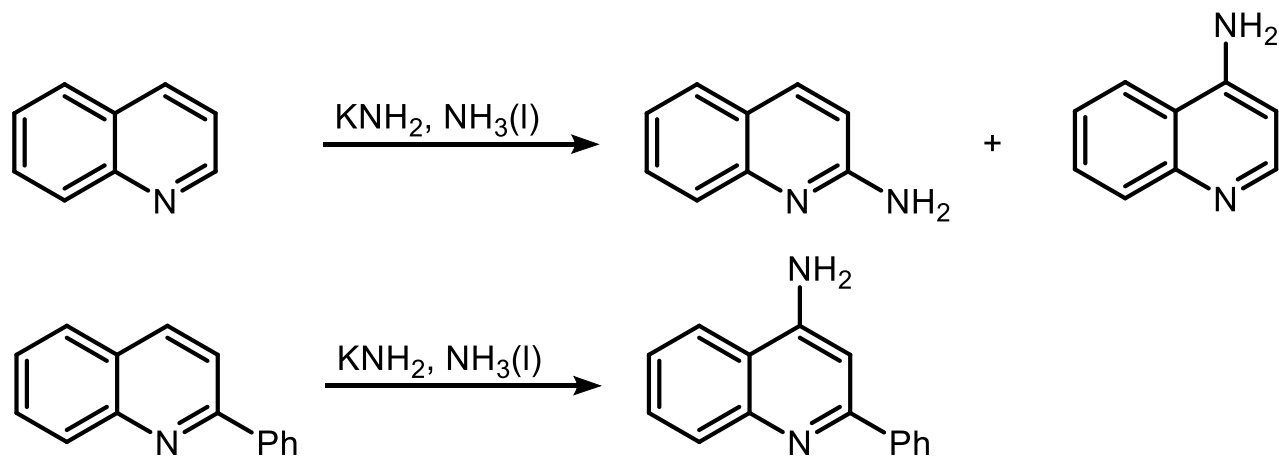


Bromierung

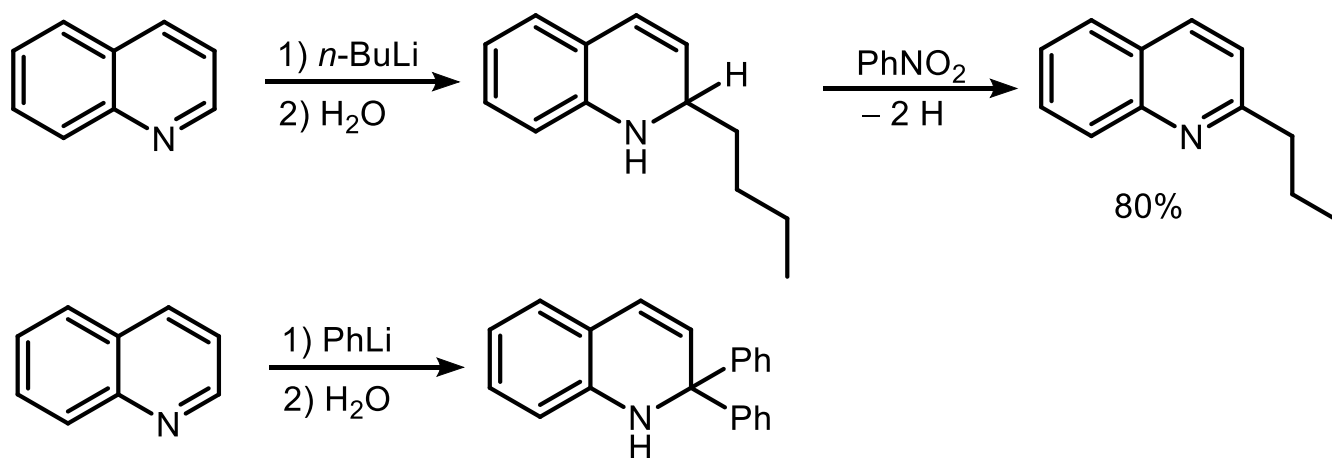


2) Nucleophile Substitutionsreaktionen (S_NAr):

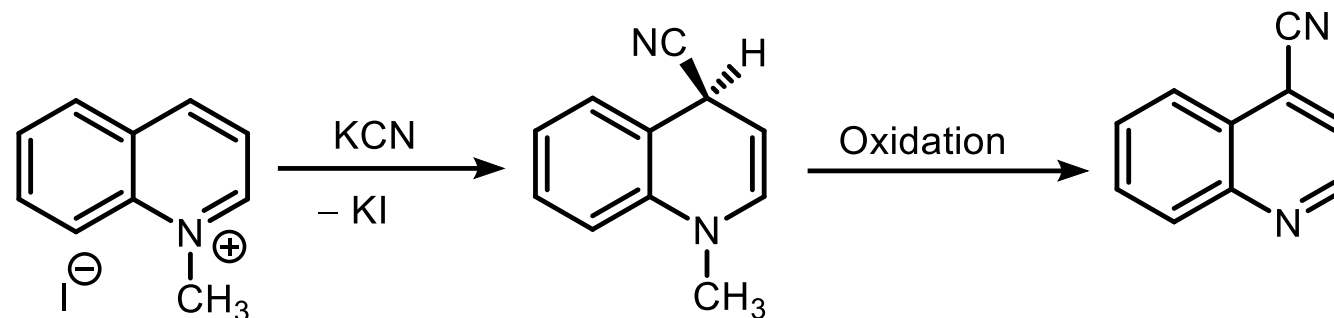
Tschitschibabin-Reaktion



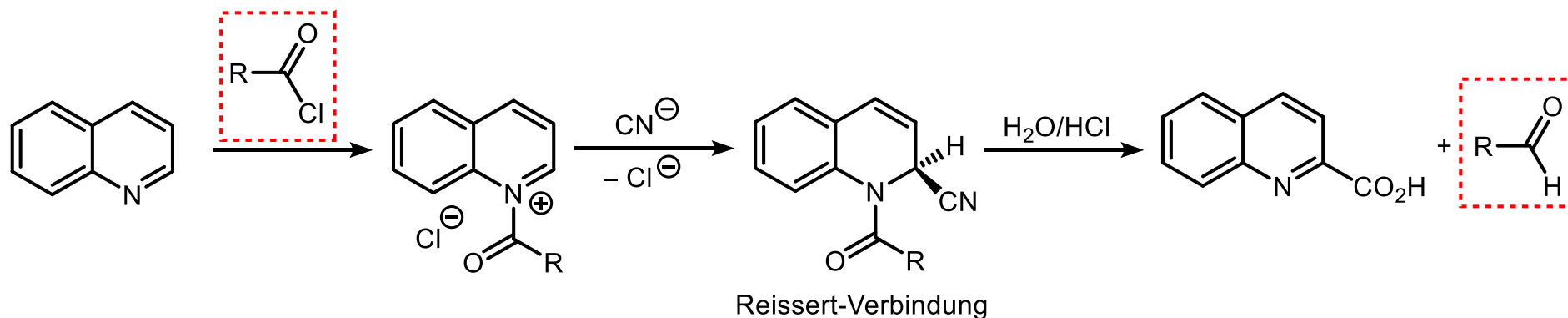
Ziegler-Reaktion



2) Nucleophile Additionen an Chinoliniumionen:

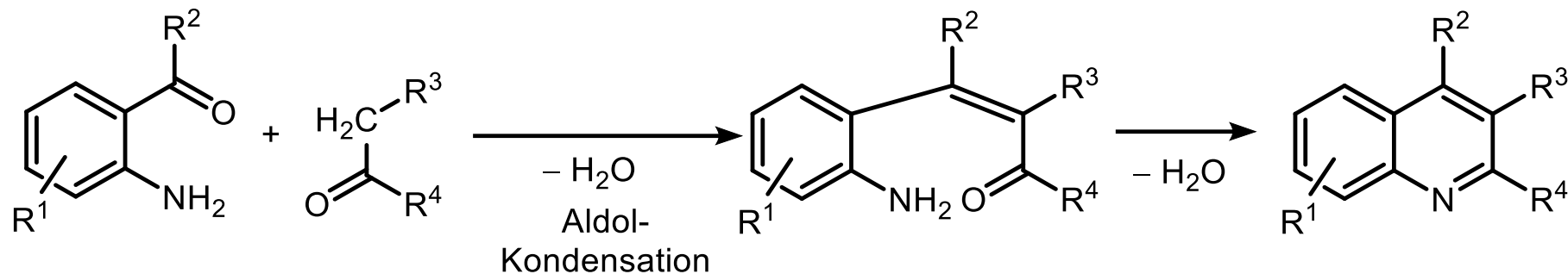


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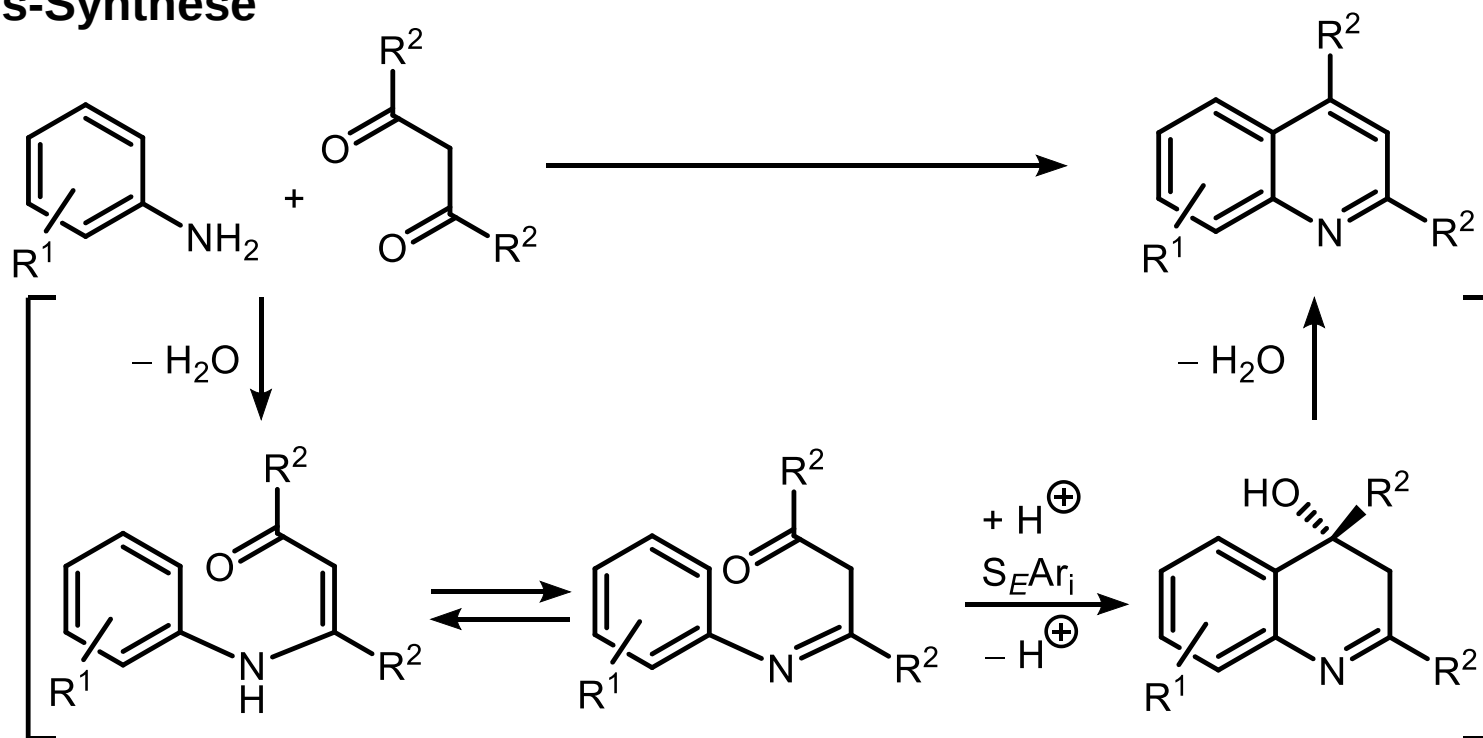


5.1.2.3. Ringaufbauende Synthesen

1) Friedländer-Synthese



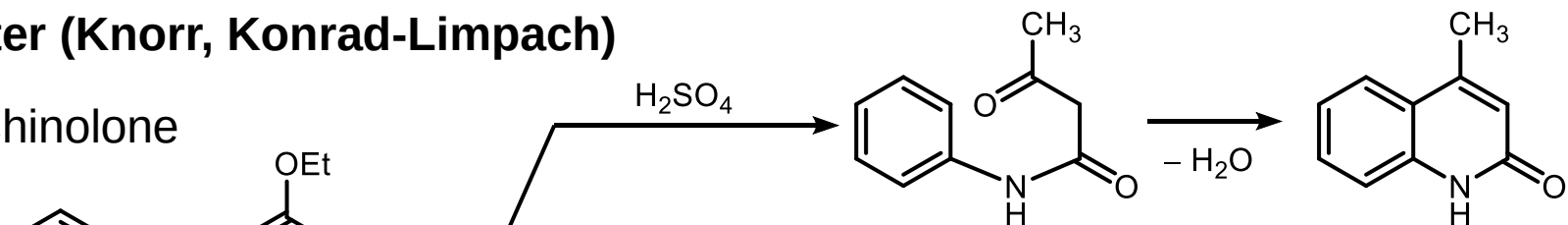
2) Combes-Synthese



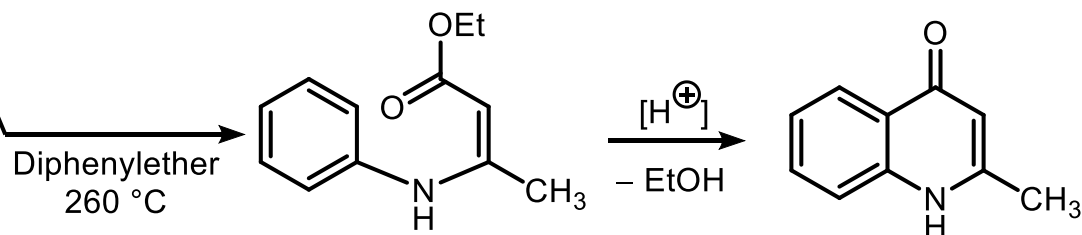
5.1.2.3. Ringaufbauende Synthesen

β -Ketoester (Knorr, Konrad-Limpach)

Knorr: 2-Chinolone

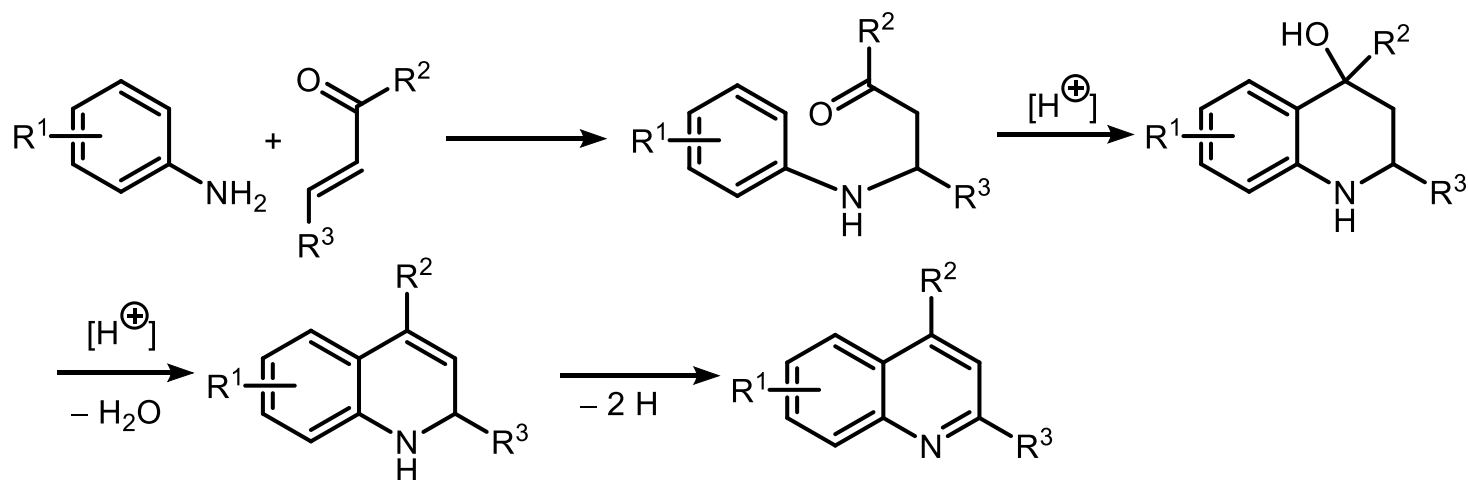


Konrad-Limpach: 4-Chinolone

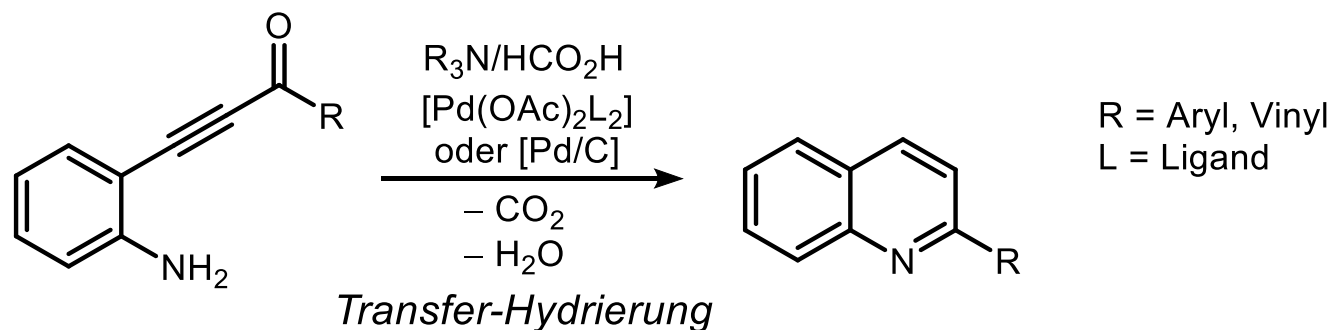


Mit Alkoxymethylenmalonester: Gould-Jacobs

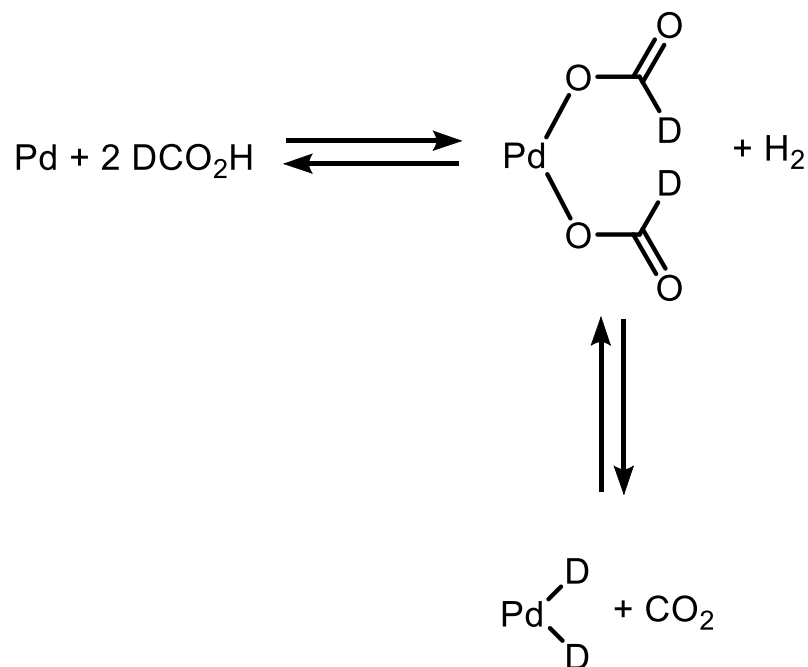
3) Skraup- (urspr. mit Glycerin) und Doebner-Miller-Synthese



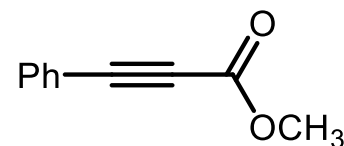
4) Marinelli-Synthese (Marinelli, *Synlett* **1999**, 401.)



Transfer-Hydrierung (Spencer, *Chem. Commun.* **1998**, 1935.)

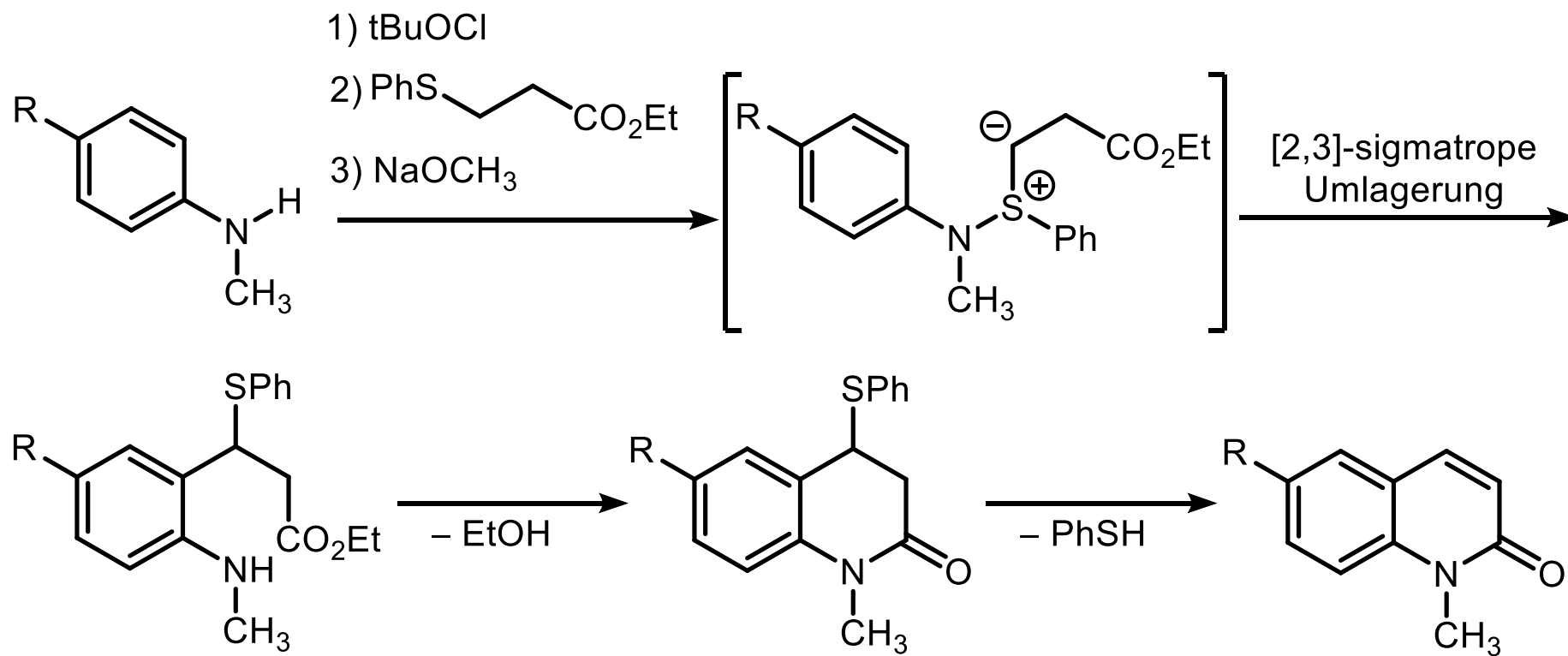


Reduktion von Phenylpropioisäuremethylester

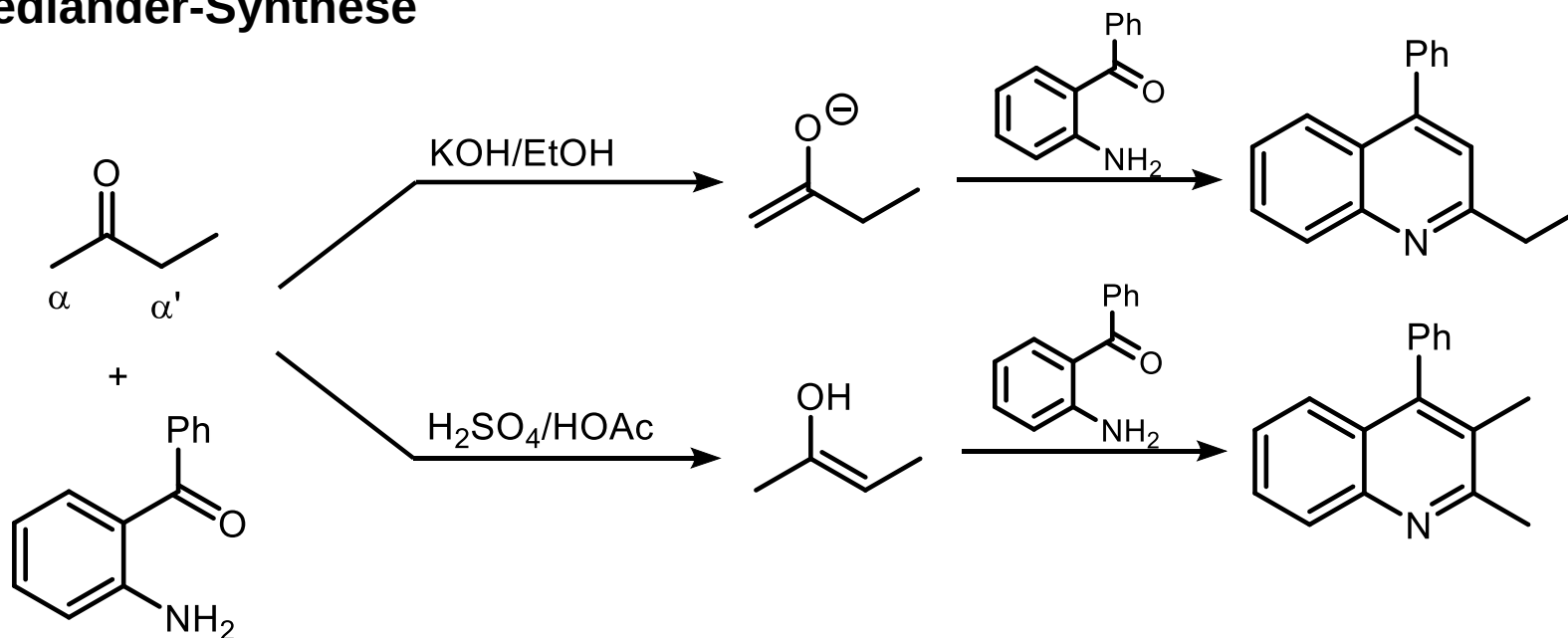


Produktverteilung [%]			
H-Quelle	H + H	H(D) + D(H)	D + D
HCO ₂ D	72	10	18
DCO ₂ H	25	18	57

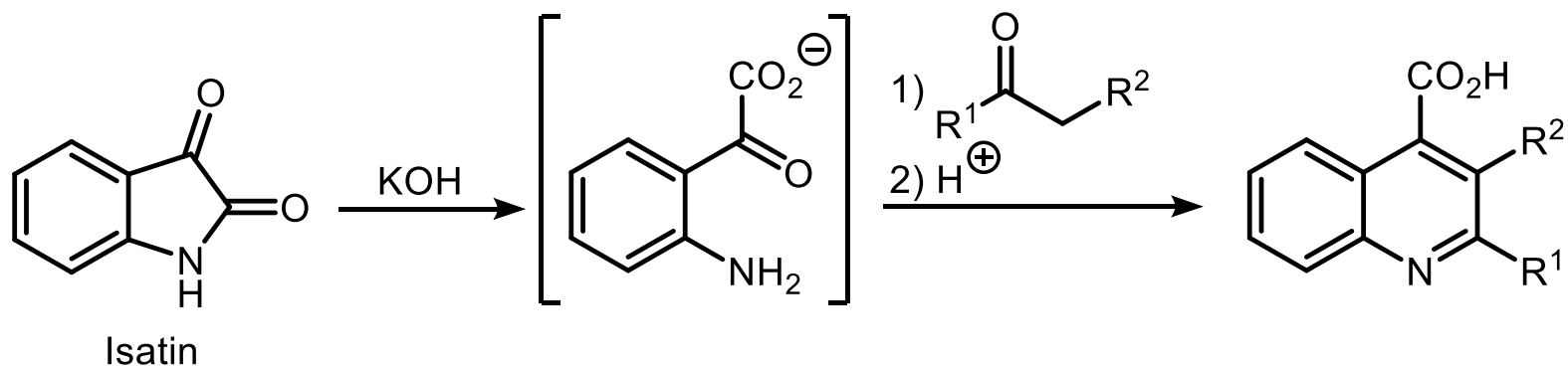
5) Gassman-Synthese (1977)



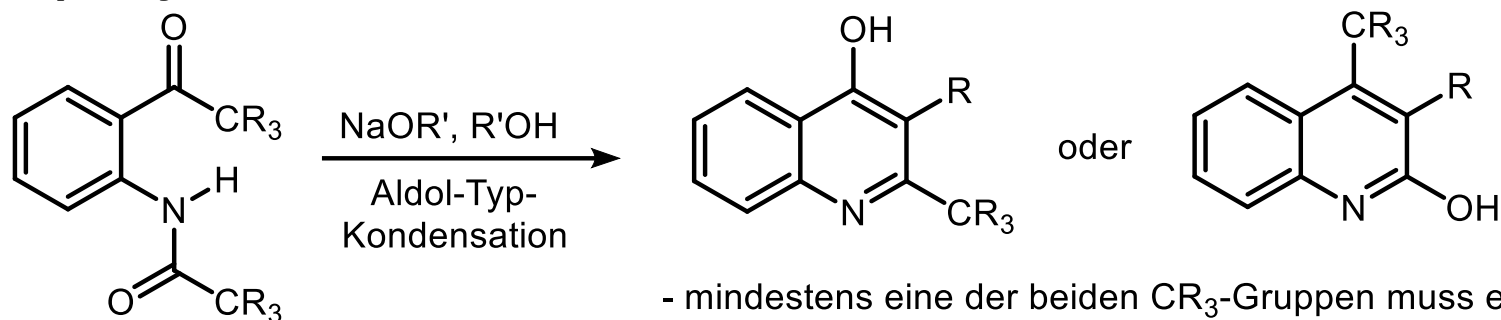
6) Friedländer-Synthese



Pfitzinger-Synthese als Spezialfall



7) Camps-Synthese



- mindestens eine der beiden CR_3 -Gruppen muss eine Methylengruppe (CH_2) sein
- je leichter eine CH-Position deprotonierbar, desto eher wird sie in den Ring inkorporiert

8) Durch formale (4+2)-Cycloaddition

